MEHLMANMEDICAL

BIOCHEMISTRY

ASSESSMENT #1

Biochemistry Assessment #1:

1. An 18-month-old girl is brought to the GP for developmental concerns. She demonstrates poor eye movements and her blood lactate levels are elevated. Audiometry reveals reduced hearing across all frequencies. Family history is remarkable for an older brother who has hypotonia and who uses a wheelchair. Her mother has required bilateral hearing aids since childhood. Which of the following best explains this patient's presentation?

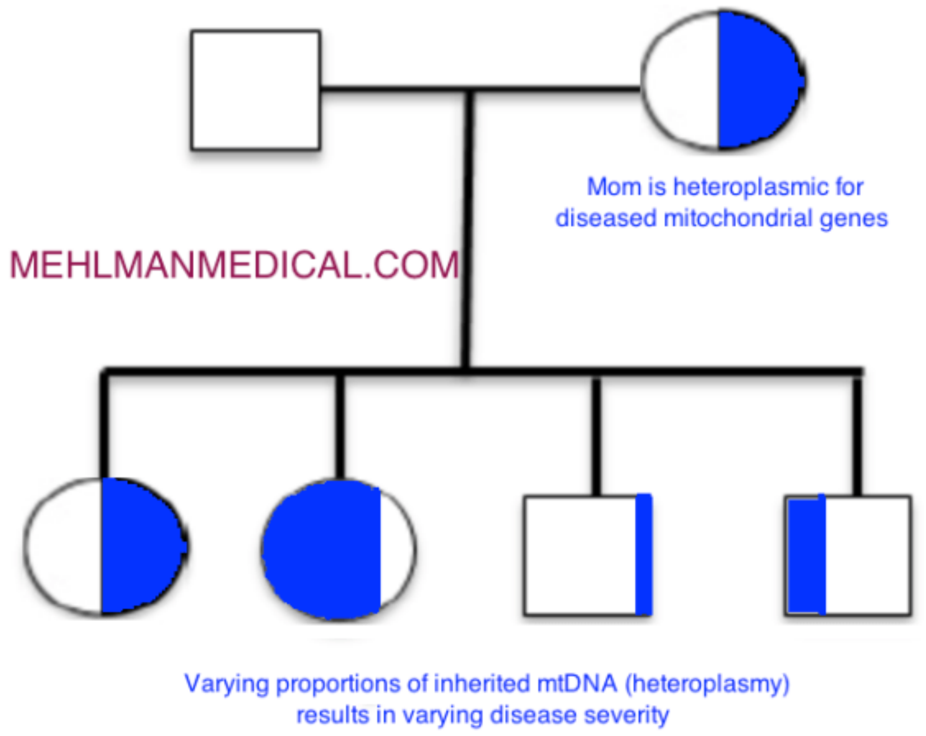
A) Autosomal recessive inheritance
B) Autosomal dominant inheritance
C) Variable expressivity
D) Incomplete penetrance
E) Heteroplasmy
F) Imprinting

The answer is E.

Heteroplasmy[1] is the presence of more than one type of mitochondrial DNA in the same individual.

This means there is a non-uniformity of mitochondrial genes inherited (i.e., cells carry different proportions of mitochondrial genes).

In turn, mitochondrial diseases frequently vary in severity among individuals because of differing ratios of diseased vs wild-type mitochondrial genes inherited.



Mitochondrial disorders are classically associated with hypotonia, reduced hearing and extraocular muscle function, and increased serum lactate.[2] Inheritance is strictly maternal, i.e., inherited through the mother only. Tissues high in mitochondrial density such as the skeletal muscles, inner ear sensory cells, and extraocular muscles are usually affected[3, 4]

Tangential but HY genetics terms:

Variable expressivity

- All individuals with diseased genotype have diseased phenotype (i.e., everyone with diseased allele shows it outwardly), but individuals merely vary in presentation.
- USMLE-favorite example is NF1, where some individuals may have, e.g., neurofibromas, cafe au lait spots, pheochromocytoma, Lisch nodules, etc., whereas other individuals might only have axillary/groin freckling. This is even when both individuals may carry the same exact mutation.

- In the case of heteroplasmy, the variable presentation among individuals is due to *different proportions* of diseased mitochondrial genes inherited.

Incomplete penetrance

- Not everyone with diseased genotype has diseased phenotype (i.e., not everyone shows it outwardly; it "skips" some people).
- An example is BRCA1/2 mutations – i.e., not everyone with the mutation will go on to develop cancer (usually breast).

Imprinting means receiving one allele from each parent but one is preferentially silenced.

Maternal imprinting means mom's allele is silenced; only dad's allele is expressed.

Paternal imprinting means dad's allele is silenced; only mom's allele is expressed.

For many genes, *it is normal* that although two copies are received (one from each parent), only one copy may be expressed due to the other being preferentially silenced (via methylation).

This is one of the reasons why two individuals of the same gender cannot produce an offspring through *in vitro* techniques, even if the combination of chromosomes is balanced.

Bottom line: Heteroplasmy is an important term to know for mitochondrial inheritance.

1) https://www.sciencedirect.com/science/article/abs/pii/S0168952597012663

2) https://academic.oup.com/brain/article/127/10/2153/404539

3) https://www.sciencedirect.com/science/article/abs/pii/S109671921530024X

4) https://science.sciencemag.org/content/242/4884/1427

2. A 23-year-old female, gravida 5, para 1, aborta 4, gives birth to a stillborn child who has multisystem organ abnormalities, limb deformities, and cleft lip. Which of the following is the most likely explanation for the presentation of the newborn?

A) Chromosomal rearrangement (unbalanced)
B) Germline mosaicism
C) Chimerism
D) Dominant lethal mutation
E) Microsatellite instability
F) Somatic mosaicism

The answer is A.

Chromosomal abnormality in the neonate is most likely based on the maternal history of recurrent miscarriage (aborta 4).

Most spontaneous abortions are due to an unbalanced chromosomal abnormality. Trisomy 16 is one of the most common causes of genomic imbalance.[1]

Germline mosaicism is when an individual (usually a male) who does not carry a mutation for a disease in somatic (body) cells, develops a mutation in a primordial germ cell, leading to some offspring with disease in all somatic (body) cells. **Achondroplasia** is a very common example.[2] It is an AD disorder, yet both parents are unaffected. This is because the mutation is germline, where it is present in some of the father's sperm but not in others.

Chimerism is when an organism's cells contain genes from more than two parents due to fusion of two zygotes. In humans, since viable chimera do not occur, this term is instead sometimes used to describe organ transplant recipients.[3]

Dominant lethal mutations are rare because they almost always lead to death of an organism before it can pass the lethal allele to offspring. Huntington disease is an example of a lethal TNR disorder in which the affected individual can still reproduce.[4]

Microsatellite instability is associated with Lynch syndrome (HNPCC), a disorder of colonic polyps, gynecologic tumors, and other soft tissue tumors.[5] The USMLE wants you to know this phenomenon is associated with **mismatch repair genes.**[6]

Somatic mosaicism is when a percentage of an individual's cells carry a different genotype or karyotype based on one cell having incurred **a post-fertilization mitotic error,** followed by continued embryogenesis.[7]

Bottom line: Chromosomal abnormalities are the most common cause of spontaneous abortion. This will almost always transpire in first trimester.

1) https://www.ncbi.nlm.nih.gov/pubmed/16232180

2) https://www.ncbi.nlm.nih.gov/pubmed/18266238

3) https://www.nejm.org/doi/full/10.1056/nejmoa074191

4) https://www.tandfonline.com/doi/full/10.4161/15384101.2014.950538

5) https://cancerres.aacrjournals.org/content/53/24/5853.short

6) https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-2559.2009.03392.x

7) https://onlinelibrary.wiley.com/doi/abs/10.1002/(SICI)1096-8628(20000306)91:1%3C39::AID-AJMG7%3E3.0.CO;2-L

3. A 36-year-old man becomes wheelchair-bound after a cliff diving accident that leaves him a quadriplegic. Over the course of several months, his muscle mass dramatically declines. Which of the following intramyocytic processes might explain this change?

A) Cori cycle
B) Polymethylation
C) Ubiquitination
D) Beta-oxidation
E) Glycolysis
F) Glycogenolysis

The answer is C.

In a state of atrophy, proteins will be tagged with ubiquitin, which targets them to the proteasome for degradation.[1]

The Cori cycle is when a muscle cell produces lactate, which then goes to the liver to be converted to glucose, which then goes back to the muscle cell to be utilized for energy.[2]

Methylation is most often associated with gene silencing.[3]

Beta-oxidation is the intramitochondrial process via which fatty acids are broken down.[4]

Glycolysis is not responsible for muscle cell breakdown in states of atrophy.

Glycogenolysis, whilst catabolic, is not responsible for loss of muscle mass during states of atrophy.

Bottom line: Ubiquitination is necessary for proteins to be targeted to the proteasome for degradation.

1) https://www.sciencedirect.com/science/article/pii/0092867494903964

2) https://www.sciencedirect.com/science/article/pii/0092867494903611

3) https://www.nature.com/articles/ncponc0354

4) https://www.sciencedirect.com/science/article/abs/pii/000527609190015A

4. A study involving pancreatic acinar cells leads to increased lipase products detectable on Western blotting, but no changes on Northern or Southern blotting. Which of the following might explain these findings?

A) Transcription factor binding
B) Release of lipase mRNA from P-bodies
C) Export of lipase mRNA from the nucleus
D) Increase in lipase gene silencing proteins

The answer is B.

mRNA is often stored on intracytosolic P-bodies until needed for translation (i.e., protein production).[1]

The USMLE merely wants you to know that P-bodies exist and that they're intracytosolic.

If mRNA is released from P-bodies, they can be translated to protein (↑ detection on Western blotting). The amount mRNA present in the cytosol should remain unchanged, and therefore Northern blotting should be unaffected.

Transcription factor binding would increase translation of DNA to RNA (↑ detection on Northern blotting), and in turn, RNA translation to protein (↑ detection on Western blotting).

Export of lipase mRNA from the nucleus would increase the RNA detectable on Northern blotting, and in turn, this RNA would be translated to protein (↑ detection on Western blotting).

Increase in lipase gene silencing proteins would decrease RNA and protein, leading to a decrease in detection on both Northern and Western blotting.

Bottom line: Southern blotting detects DNA; Northern blotting detects RNA; Western blotting detects protein. P-bodies are intracytoplasmic "docking stations" through which mRNAs can be bound and stored for future use.

1) https://www.ncbi.nlm.nih.gov/pubmed/28965817

5. A 2-year-old boy is brought to the GP by his parents because they are concerned about writhing movements he makes during sleep. Physical examination shows a small nodule growing from one of his nailbeds and small peach-colored papules adjacent his nose bilaterally. He has a few hyperpigmented velvety lesions on his trunk. Which of the following is most likely to be seen in this patient?

A) Perioral melanosis
B) Congenital bilateral cataracts
C) Pulmonic valve stenosis
D) Cardiac rhabdomyoma
E) Renal cell carcinoma

The answer is D.

Tuberous sclerosis is a HY autosomal dominant disorder characterized by periventricular nodules (leading to seizure, as this child has had during sleep), adenoma sebaceum (angiofibromas found in a butterfly distribution on the face), subungual fibromas (nailbed tumors), cardiac rhabdomyoma, and renal angiomyolipoma, among other findings.[1]

Perioral melanosis (hyperpigmentation around the mouth) is seen classically in Peutz-Jeghers syndrome, a disorder characterized by hamartomatous colonic polyps.[2]

Congenital bilateral cataracts can be seen in neurofibromatosis type II (NF2).[3]

Pulmonic valve stenosis is classic for DiGeorge syndrome secondary to tetralogy of Fallot.[4] It is also classic for Noonan syndrome.[5] You will not get asked about Noonan syndrome on the USMLE Step 1, but DiGeorge is exceedingly HY.

Renal cell carcinoma (RCC; sometimes bilateral) is characteristic of von Hippel-Lindau (VHL), a disorder characterized by cerebellar and retinal hemangioblastomas and RCC.[6]

Bottom line: Tuberous sclerosis is an AD disorder with very characteristic features that must be memorized for the USMLE.

1) https://www.nejm.org/doi/full/10.1056/NEJMra055323

2) https://www.nejm.org/doi/full/10.1056/NEJM198706113162404

3) https://www.thieme-connect.com/products/ejournals/html/10.1055/s-2005-837581

4) https://link.springer.com/article/10.1007/BF00443218

5) https://adc.bmj.com/content/67/2/178.short

6) https://www.nature.com/articles/ng0594-85

6. A 2-month-old girl is evaluated for a rare congenital syndrome characterized by decreased vesicular shuttling to the Golgi apparatus. Which of the following would most likely be seen on electron microscopy of this patient's cells?

A) Distended peroxisomes
B) Polyubiquitination
C) Dilated smooth endoplasmic reticulum
D) Enlargement of rough endoplasmic reticulum
E) Increased nuclear to cytoplasm ratio

The answer is D.

Shuttling from the RER to the Golgi occurs via COP II.[1] Defective shuttling would lead to enlargement of the RER, where proteins would be sequestered.[2]

Peroxisomes are responsible for very long-chain fatty acid and branched-chain fatty acid metabolism. Disorders include Refsum disease, adrenoleukodystrophy, and Zellweger syndrome.[3] The USMLE's assessment of peroxisomal disorders is very rare. But you should at least be aware of the mere function of the peroxisome, and perhaps keep these disease names on the far back-burner of your mind as being peroxisomal.

Ubiquitination is necessary for targeting of intracellular protein to the proteasome for degradation.[4]

The smooth endoplasmic reticulum is necessary for steroid synthesis and lipid peroxidation.[5]

An increased nuclear to cytoplasm ratio is seen in many cancers.[6]

Bottom line: Many proteins are produced at the RER and then shuttled to the Golgi apparatus.

1) https://www.sciencedirect.com/science/article/pii/S0092867400803797

2) https://www.fasebj.org/doi/abs/10.1096/fj.03-1210fje

3) https://www.sciencedirect.com/science/article/pii/S0167488906002874

4) https://www.sciencedirect.com/science/article/pii/0092867494903964

5) https://nyaspubs.onlinelibrary.wiley.com/doi/abs/10.1111/j.1749-6632.1965.tb12247.x

6) https://clincancerres.aacrjournals.org/content/14/9/2681.short

7. Two of a couple's three children have tuberous sclerosis. Analysis of the father's sperm reveals 1 in 5 carries an in-frame deletion involving amino acids 1746_1751 (HIKRLR; c.5238_5255del) in exon 40 of the *TSC2* gene. Which of the following best explains this finding?

A) Germline mosaicism
B) De novo mutation
C) Imprinting
D) Advanced paternal age
E) Somatic mosaicism

The answer is A.

Germline mosaicism is when an individual (usually a male) who does not carry a mutation for a disease in somatic (body) cells, develops a mutation in a primordial germ cell, leading to some offspring with disease in all somatic (body) cells.

Achondroplasia is a very common example[1], but this may also be seen in TSC.[2] It is an AD disorder, yet both parents are not affected. This is because the mutation is germline, where it is present in some of the father's sperm but not in others.

The in-frame deletion involving amino acids 1746_1751 (HIKRLR; c.5238_5255del) in exon 40 of the *TSC2* gene is a common mutation in tuberous sclerosis. You will **not** get asked this on the USMLE, but nevertheless explains the relevance to this question.[3]

De novo mutations are common in autism spectrum disorder (ASD), which bears significant locus heterogeneity.[4] An individual may acquire a new mutation at any one of a number of loci, resulting in ASD.

Imprinting means receiving one allele from each parent but one is preferentially silenced.

Advanced paternal age is associated with increased risk of autism spectrum disorder (ASD).[5]

Somatic mosaicism is when a percentage of an individual's cells carry a different genotype or karyotype based on one cell having incurred **a post-fertilization mitotic error,** followed by continued embryogenesis.[6]

Bottom line: Germline mosaicism is an important USMLE-tested mechanism in which a parent who's genotypically normal in somatic (body) cells (and in turn, phenotypically normal) may have recurrent offspring with an AD disorder.

1) https://www.ncbi.nlm.nih.gov/pubmed/18266238

2) https://jamanetwork.com/journals/jamaneurology/article-abstract/776480

3) https://www.nature.com/articles/gim200715

4) https://www.nature.com/articles/ng.3050

5) https://www.cambridge.org/core/journals/the-british-journal-of-psychiatry/article/paternal-age-at-birth-and-highfunctioning-autisticspectrum-disorder-in-offspring/87C33805E207715596B08C9DF529D9FC

6) https://onlinelibrary.wiley.com/doi/abs/10.1002/(SICI)1096-8628(20000306)91:1%3C39::AID-AJMG7%3E3.0.CO;2-L

8. A recent outbreak of H7N3 high-pathogenic avian influenza (HPAI) occurring in Venezuela has lead to many deaths. Sequencing of all eight genes of the HPAI virus and other strains of influenza viruses demonstrates minor differences except at the hemagglutinin cleavage site. The HPAI isolates bear a 30-nucleotide insertion. Which of the following mechanisms might explain this finding?

A) Viral reassortment
B) Viral recombination
C) Frameshift mutation
D) Transposon activity
E) Specialized viral translocation

The answer is A.

The process described is **antigenic shift,** in which two or more influenza strains create a novel virus via **reassortment** of segments.[1] The resulting novel virus is often able to cause **pandemics,** or widespread death, due to the human immune system's difficulty recognizing such a new influenza strain.[2]

Antigenic shift is frequently contrasted with **antigenic drift,** which causes less severe epidemics, often due to mere point mutations in hemagglutinin or neuraminidase.[3, 4]

Viral recombination is when two viruses coinfect a host cell and then yield progeny with some genes from each virus. Whereas any gene combination may occur with recombination, in the case of reassortment, there are *whole segments* that are exchanged between segmented viruses.[5, 6]

Frameshift mutations cause an abnormally truncated protein due to an abrupt shift in the mRNA reading frame via an insertion or deletion of 1 or 2bp.[7]

Transposons involve genetic sequences that move across the genome and reinsert.[8]

Specialized viral translocation is a distractor term synthesized for this question.

Bottom line: Reassortment of segments between an influenza strain and a second virus is referred to as antigenic shift and results in a highly novel virus capable of causing widespread pandemics. Antigenic drift refers usually to point mutations in hemagglutinin or neuraminidase and results in less severe epidemics.

1) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3323103/

2) https://academic.oup.com/jac/article/44/suppl_2/3/2473569

3) https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-2958.1994.tb00394.x

4) https://jvi.asm.org/content/48/1/52.short

5) https://www.sciencedirect.com/science/article/pii/S1879625718300646

6) https://royalsocietypublishing.org/doi/abs/10.1098/rspb.1996.0231

7) http://symposium.cshlp.org/content/31/77.extract

9. Increased c-jun protein expression has been associated with various cancers in humans, notably the MCF-7 breast cancer cell line. It has been found that phosphorylation of c-Jun at serines 63 and 73, as well as at threonines 91 and 93, leads to upregulation of c-jun target genes. Which of the following best describes c-Jun?

A) Transcription factor
B) Anti-apoptotic molecule
C) Tyrosine kinase
D) Serine-threonine kinase

The answer is A.

c-Jun is a transcription factor.[1] Its regulation may in part be mediated by serine-threonine kinases, as the question stem implies.[2]

Bcl-2 is an extremely HY anti-apoptotic molecule for the USMLE. It is upregulated in follicular lymphoma secondary to a t(14;18) translocation.[3, 4]

Bcr-abl oncogenic tyrosine kinase is HY for the USMLE. It is a result of the Philadelphia chromosome, a t(9;22) translocation.[5]

Bottom line: c-Jun is a transcription factor.

1) https://www.nature.com/articles/1202989

2) https://www.nature.com/articles/nature03914

3) https://science.sciencemag.org/content/228/4706/1440

4) http://www.jbc.org/content/272/18/11671.short

5) https://doi.org/10.1182/blood-2008-07-077958

10. A female newborn who has blue irides and talc-white skin is diagnosed with oculocutaneous albinism. The most likely cause may be which of the following?

A) Defective shuttling of tyrosinase in the rough endoplasmic reticulum of lysosomes
B) Defective shuttling of melanin within the rough endoplasmic reticulum of lysosomes
C) Defective melanin packaging and exocytic vesicle formation
D) Defective shuttling of tyrosinase in the rough endoplasmic reticulum to melanosomes
E) Defective shuttling of tyrosine hydroxylase in the rough endoplasmic reticulum of lysosomes

The answer is D.

Oculocutaneous albinism is caused by a deficiency of the enzyme **tyrosinase** produced by **melanosomes.**[1]

The mechanism may be due to disruption of tyrosinase trafficking within the rough endoplasmic reticulum.[2]

Bottom line: For oculocutaneous albinism, the enzyme deficiency the USMLE wants is **tyrosinase,** and the organelle is the **melanosome.**

1) https://www.sciencedirect.com/science/article/pii/S0022202X94916160

2) https://jcs.biologists.org/content/116/15/3203.short

11. An 8-year-old boy with a history of respiratory tract infections has a sweat chloride test of 87 mEq/L (normal <60). He presents with a fever of 102F and mild right-sided chest pain. Chest x-ray shows right lower lobe infiltrates and a small pleural effusion. What is the most likely pathogen causing this patient's condition?

A) *Streptococcus pneumoniae*
B) *Listeria monocytogenes*
C) *Legionella pneumophila*
D) *Pneumocystis jirovecii*
E) *Staphylococcus aureus*
F) *Pseudomonas aeruginosa*

The answer is E.

Cystic fibrosis is assessed under the envelope of biochemistry for USMLE Step 1.

During the first decade of life, *S. aureus* and non-typable *Haemophilus influenzae* eclipse *Pseudomonas* as the most common cause of pneumonia in CF patients. In the second decade onward, *Pseudomonas* eclipses *S. aureus*.[1, 2, 3]

This is exceedingly HY on the USMLE and a question students frequently get wrong.

Bottom line: If the patient is in the first decade of life, the answer is *S. aureus,* **not** *Pseudomonas,* as the most likely causal organism for pneumonia in CF.

1) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2586015/

2) https://www.sciencedirect.com/science/article/pii/S1569199307000987

3) https://europepmc.org/abstract/med/11891518

12. A 16-year-old boy with chronic lung disease managed with dornase-alfa and guaifenesin presents with diminished reflexes, visual impairment, pigmented retinopathy, weakness, and ataxia. Which of the following is the most likely explanation for these findings?

A) Drug-induced neuropathy
B) Exocrine pancreas insufficiency
C) Reactivation and progression of latent infection
D) Anti-neuronal antibodies

The answer is B.

Vitamin E deficiency may be seen in cystic fibrosis secondary to fat-soluble vitamin malabsorption due to exocrine pancreas insufficiency.[1]

Vitamin E deficiency may present with ataxia, pigmented retinopathy, visual impairment, ataxia, and weakness.[2]

The USMLE is obsessed with you making the link between diseases that cause intestinal malabsorption and fat-soluble vitamin deficiencies. You may instead get a vignette that describes, e.g., bowing of the tibias, which would be vitamin D deficiency. Get the point?

Dornase-alfa is a nucleotidase mucolytic used in the management of CF.[3]

Guaifenesin is an expectorant. Its utility is debatable in CF.[4]

Bottom line: CF causes fat-soluble vitamin malabsorption resulting in their respective presentations of deficiency.

1) https://www.sciencedirect.com/science/article/abs/pii/S000282230800179X

2) https://n.neurology.org/content/36/7/917.short

3) https://www.ncbi.nlm.nih.gov/pubmed/27043279

4) https://www.ncbi.nlm.nih.gov/pubmed/17594730

13. A 2-year-old boy with coarse facial features, gingival hypertrophy, thick tongue, short neck and thoracic cage, and cardiomegaly is found to have a deficiency of an enzyme called N-acetylglucosamine-1-phosphotransferase. He has markedly increased activity of plasma acid hydrolases. The process that is defective in this patient occurs in which of the following organelles?

A) Mitochondria
B) Rough endoplasmic reticulum
C) Smooth endoplasmic reticulum
D) Lysosome
E) Peroxisome
F) Golgi apparatus

The answer is F.

This boy has **I-cell disease,** which is a lysosomal storage disease caused by deficiency of N-acetylglucosamine-1-phosphotransferase.[1]

This leads to a failure **at the Golgi** to target lysosomal enzymes to the lysosomes.[2, 3] In order for this targeting to occur, lysosomal enzymes must be tagged with mannose-6-phosphate (M6P) at the Golgi apparatus.[4]

The result is the lysosomal enzymes are secreted extracellularly, leading to higher levels in the plasma.[5]

Features of the disease notably include coarse facial features, gingival hypertrophy, thick tongue, short neck and thoracic cage, and cardiomegaly.[6]

Bottom line: I-cell disease is a failure of the **Golgi** to tag lysosomal enzymes with **mannose-6-phosphate.** This results in their secretion into the plasma.

1) https://www.sciencedirect.com/science/article/pii/0006291X82910762

2) https://www.sciencedirect.com/science/article/pii/009286748490223X

3) https://www.sciencedirect.com/science/article/pii/0006291X72903105

4) http://jcb.rupress.org/content/123/1/99.abstract

5) https://www.translationalres.com/article/0022-2143(74)90107-3/fulltext

6) https://link.springer.com/article/10.1007/BF00569920

14. A 25-year-old man goes to Bermuda for a 5-day vacation in April. He develops four watery stools per day for three days. What is the most likely mechanism of the presentation described?

A) ↑ cAMP
B) ↑ GTPase activity
C) ↑ Diacylglycerol
D) ↓ cAMP
E) ↓ Diacylglycerol

The answer is A.

Travelers diarrhea is due to enterotoxigenic *E. coli* (ETEC) heat-labile or -stable toxins.[1]

This usually results in a watery diarrhea after travel to lower income areas outside the United States.[1, 2]

Heat-labile toxin ADP ribosylates adenylyl cyclase, causing increased G-alpha-s activity via inhibiting GTPase activity (normally GTPase activity shuts off G-proteins). **This leads to increased cAMP levels.**[3]

Heat-stable toxin ADP ribosylates guanylyl cyclase. **This leads to increased cGMP levels.**[4]

Bottom line: The USMLE wants you to know that ETEC HL toxin = ↑cAMP levels; ETEC HS toxin = ↑ cGMP levels. *Vibrio cholera* toxin has the same mechanism as HL toxin[5]; *Yersinia enterocolitica t*oxin has the same mechanism as HS toxin.[6]

1) http://article.sapub.org/10.5923.j.ajmms.20140405.03.html

2) https://www.ncbi.nlm.nih.gov/books/NBK7710/

3) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1364263/

4) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3153297/

5) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3089050/

6) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC414498/

15. An 8-year-old boy has a history of multiple fractures, progressive skeletal deformity, scoliosis, and poor longitudinal growth. He falls below the third percentile for height. His mother has a history of several miscarriages and he does not have any siblings. This patient's pathology might be best explained by which of the following?

A) Decreased alanine content in elastin
B) Defective fibrillin sheathing of elastin
C) Deficiency of carbonic anhydrase II
D) Defective collagen I hydrogen bonding
E) Antibodies against the alpha-3 chains of type IV collagen

The answer is D.

Osteogenesis imperfecta (OI) is due to defective collagen I.[1]

There are many types of OI.

Type I is classic and presents with blue sclerae, fractures, and hearing impairment. Type II is often lethal *in utero*. Type III presents as fractures, progressive skeletal deformity, scoliosis, and poor longitudinal growth.[2]

It is generally accepted that defective hydrogen bonding is one of the salient mechanisms through which collagen I may be weakened in various types of OI.[3]

Marfan syndrome is a mutation in the fibrillin genes *FBN1* and *FBN2*. Fibrillin is a glycoprotein that forms a sheath around elastin.[4]

Deficiency of carbonic anhydrase II causes osteopetrosis.[5]

Antibodies against the alpha-3 chains of type IV collagen cause Goodpasture syndrome.[6]

Bottom line: Osteogenesis imperfecta is due to defective collagen I. Impaired hydrogen bonding is a salient mechanism for weakened collagen I structure. The USMLE will sometimes mention that the patient's mother has a Hx of multiple miscarriages.

1) https://www.ncbi.nlm.nih.gov/pubmed/12362985

2) https://www.medigraphic.com/pdfs/opediatria/op-2018/op182e.pdf

3) https://pubs.acs.org/doi/abs/10.1021/bi035676w

4) https://www.ncbi.nlm.nih.gov/pubmed/7493032

5) https://www.ncbi.nlm.nih.gov/pubmed/15300855

6) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC44166/

16. A 26-year-old woman comes to the physician because of a two-month history of severe muscle cramping and pain. She recently started a kickboxing class but says she cannot finish the class because of the pain. Serum creatine kinase is elevated. Post-exercise venous sampling reveals no increase in lactic acid concentration.

Urine dipstick: 1+ blood
Urine light microscopy: 0-1 RBCs per high-power field

What is the mechanism for this patient's disease?

A) Deficiency of glucose-6-phosphatase
B) Deficiency of glucose-6-phosphate dehydrogenase
C) Deficiency of alpha-1-4-glucosidase (lysosomal acid maltase)
D) Deficiency of alpha-1-6-glucosidase
E) Deficiency of myophosphorylase

The answer is E.

The diagnosis is McArdle syndrome (glycogen storage disease type V). The deficiency is muscle glycogen phosphorylase (myophosphorylase).

Creatine kinase levels increase with intense exercise, suggestive of muscle breakdown[1], however blood lactate levels do not rise abnormally.[2]

Rhabdomyolysis may be seen and can cause renal failure with acute tubular necrosis, since myoglobin is nephrotoxic.[3] However rhabdomyolysis can also present as a false (+) blood on urine dipstick – i.e., no or few RBCs on LM despite + blood on dipstick. This is because the dipstick cannot differentiate between myoglobinuria and actual hemoglobin in RBCs.[4]

Glucose-6-phosphatase deficiency causes von Gierke disease (glycogen storage disease type I).[5]

Deficiency of alpha-1-4-glucosidase (lysosomal acid maltase) causes Pompe disease (GSD type II).[6]

Deficiency of alpha-1-6-glucosidase causes Cori syndrome (GSD type III).[7]

Bottom line: The glycogen storage diseases are HY for the USMLE. All are fair game. McArdle (type V) most often presents as an adult who has severe cramping and pain with intense exercise.

1) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4009758/

2) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2496477/

3) https://www.ncbi.nlm.nih.gov/pubmed/17383055

4) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3395513/

5) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3118311/

6) https://www.ncbi.nlm.nih.gov/pubmed/11949932

7) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2949315/

17. A 22-year-old male has a two-day history of tenderness in his right axilla and burning pain in his right hand. His temperature is 101.5F. Examination of the hand shows a small crop of vesicles on the distal aspect of the third digit. The most appropriate treatment would require inhibition of which of the following?

A) RNA polymerase
B) DNA polymerase
C) Protease
D) Dihydrofolate reductase
E) Purine synthesis

The answer is B.

Herpetic whitlow is a herpes (HSV1 or HSV2) infection of a digit that presents as a vesicular eruption with axillary lymphadenopathy and burning or stabbing pain (herpetic neuralgia).[1] It is classically seen in those who work in oral hygiene (i.e., dentists, hygienists, etc.).[2]

Treatment is acyclovir, which is a DNA polymerase inhibitor.[3] This will sometimes present in an answer choice on the USMLE as a drug that causes “chain termination.”

Ribavirin is a HY RNA polymerase inhibitor that has utility in hepatitis C and (theoretical use in) respiratory syncytial virus infections.[4] I write “theoretical” because on the USMLE it is never the answer for RSV (always choose supportive care). Ribavirin causes hemolytic anemia.[5]

Protease inhibitors, such as ritonavir, are HY for HIV management. They cause fat redistribution and hyperglycemia.[6]

Methotrexate, trimethoprim, and pyrimethamine are HY DHF reductase inhibitors.[7] Methotrexate causes pulmonary fibrosis, hepatotoxicity, and mucositis (mouth ulcers).[8]

6-mercaptopurine is a notable purine synthesis inhibitor. It has uitlity in acute lymphoblastic leukemia (ALL).[9]

Bottom line: Acyclovir is a DNA polymerase inhibitor used to treat HSV1/2 infections (and sometimes varicella). Herpetic whitlow is herpes of a digit and is frequently seen in those who work in oral hygiene.

1) https://www.ncbi.nlm.nih.gov/books/NBK482379/

2) https://www.ncbi.nlm.nih.gov/pubmed/6957456

3) https://www.ncbi.nlm.nih.gov/pubmed/6285735

4) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6271563/

5) https://www.ncbi.nlm.nih.gov/pubmed/17168855

6) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3166345/

7) https://www.ncbi.nlm.nih.gov/pubmed/17346178

8) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5175096/

9) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4222610/

18. A 44-year-old man attends a holiday party at which he consumes a large amount of alcohol. He develops a headache the next day that he takes acetaminophen to treat. Which of the following might explain his increased risk for hepatocellular injury?

A) ↑ $NAD^+/NADH$ ratio
B) ↑ $NADH/NAD^+$ ratio
C) Inhibition of P-450 enzymes
D) Induction of P-450 enzymes

The answer is D.

Acetaminophen is metabolized hepatically into a toxic metabolite called N-acetyl-p-benzoquinone imine (NAPQI) well-known to cause acute liver failure.[1]

The interaction between acetaminophen and ethanol has long been the subject of scientific investigation (i.e., does acute vs chronic ingestion matter?)[2], but for the sake of the USMLE, what we can say is this:

Acetaminophen + alcohol *taken together* (i.e., while acutely intoxicated) → inhibition of P-450 enzymes → acute ↓ production of NAPQI (slight protective effect).[3]

Acetaminophen *taken after* sufficient time is allowed for ethanol to be metabolized out of one's system (i.e., not acutely intoxicated but recently was) → induction of P-450 enzymes → acute ↑ production of NAPQI (harmful effect). This effect is believed to be further enhanced in chronic alcohol users due to MEOS (microsomal ethanol oxidizing system; CYP2E1) upregulation. [3]

Ethanol increases the $NADH/NAD^+$ ratio, however this is not the mechanism for hepatocellular injury in conjunction with acetaminophen use.

Bottom line: Acetaminophen taken after ethanol has been acutely metabolized out of one's system increases the risk for liver injury due to induction of MEOS CYP2E1. These enzymes are acutely inhibited when acetaminophen is taken *concurrent to* ethanol consumption.

1) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3044203/

2) https://www.ncbi.nlm.nih.gov/pubmed/11776481

3) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2014937/

19. A 48-year-old woman had an esophagectomy performed three years ago for adenocarcinoma secondary to longstanding history of gastroesophageal reflux disease. She presents with a 3-month history of 15kg (33lb) weight loss and severe fatigue. Which of the following is most likely to be observably increased on electron microscopy of a deltoid muscle cell?

A) Lipofuscin
B) Increased nuclear to cytoplasm ratio
C) Autophagic vacuoles
D) Golgi
E) Centrioles

The answer is C.

Cachexia (weight loss due to cancer) is associated with increased intracellular **autophagic vacuole** formation.[1] That is, the cell is literally eating itself, or "self-eating."[2]

Lipofuscin is yellow-brown in color and is known as the "aging pigment." It accumulates in the lysosomes of aging cells and is due to age-related enhancement of autophagocytosis, a decline in intralysosomal degradation, and/or a decrease in exocytosis.[3] It is not related to cachexia.

An increased nuclear to cytoplasm ratio is seen in many cancers, but this pathologic occurrence in and of itself is not directly tied to cachexia.[4]

Centrioles are an organelle that facilitate microtubule recruitment for cell division and are unrelated to cachexia.[5]

Bottom line: Increased numbers of autophagic vacuoles are seen in the skeletal muscle of patients with cachexia.

1) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4962093/

2) https://www.ncbi.nlm.nih.gov/pubmed/30779005

3) https://www.ncbi.nlm.nih.gov/pubmed/9531959

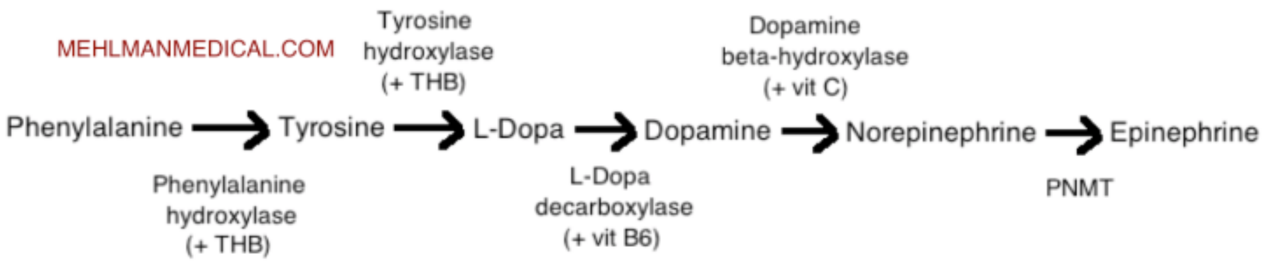
4) https://onlinelibrary.wiley.com/doi/full/10.1002/cncy.21585

5) https://www.sciencedirect.com/science/article/pii/S0960982201002895

20. An experimental animal is designed to have a defect in amino acid metabolism. Serum concentrations of phenylalanine and tyrosine are increased. Neuronal analysis reveals decreased production of dopamine and serotonin. A deficiency of which of the following might be expected?

A) Phenylalanine hydroxylase
B) Tyrosine hydroxylase
C) L-Dopa decarboxylase
D) Tetrahydrobiopterin

The answer is D.



Tetrahydrobiopterin (THB) is a cofactor for both phenylalanine hydroxylase and tyrosine hydroxylase[1], meaning that deficiency would increase both phenylalanine and tyrosine levels.

Although tyrosine hydroxylase deficiency could explain an increase in phenylalanine and tyrosine, and a decrease in dopamine, it would not explain a deficiency of serotonin. This is because serotonin synthesis (which also requires THB) starts with tryotophan, not phenylalanine.[2]

It should also be noted tangentially that whilst phenylketonuria (PKU) is almost always due to deficiency of phenylalanine hydroxylase, *malignant PKU* is due to deficiency of THB.[3]

If the USMLE chooses to ask about malignant PKU versus "regular" PKU, the type of question they'd ask would be akin to this one, and you'd have to more or less just reason your way through it.

Bottom line: The USMLE wants you to be aware of THB. Sounds like a weird and low-yield molecule, but it shows up. That's your heads up!

1) https://www.researchgate.net/publication/231210127

2) https://www.ncbi.nlm.nih.gov/pubmed/4004257

3) https://www.ncbi.nlm.nih.gov/pubmed/26817292

MEHLMANMEDICAL

BIOCHEMISTRY

ASSESSMENT #1